

POWERFACTORY

PowerFactory 2024

Technical Reference

DIgSILENT Library - Synchronous Generator Power Plants

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document offers an overview of control models for Synchronous Generator Power Plants provided in the global *PowerFactory* library, and indicates for each model its *PowerFactory* library location and the corresponding reference documents. Models are grouped per model type (e.g., AVR+exciter, power system stabilizers, etc.). Generic guidelines for common applications are mentioned at the end of the document.

Note: Models included in CGMES Profile 2.4.15 belong to the former version of the newest CGMES Profile 3.1.

Note: Models included in the DIgSILENT library are meant to be used for network equivalent purposes concerning the network reduction tool and, therefore, should not be used for general modelling purposes.

2 Synchronous Generator Power Plant Models

2.1 Governors and Turbine/Engine Models

Type Name	Description	Library	Reference
gov_PSSE_BBGOV1	European Governor Model	PSS/E	Ref: [7], Sec. 7.1
gov_IEEE_CIGRE_GT	CIGRE gas turbine-governor model	IEEE	Ref: [3], Sec. 3.1.2.4
gov_CGMES_3_1_CIGREGT	CIGRE gas turbine-governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.15 and [3], Sec. 3.1.2.4
gov_CONSTP_LFSMO	Constant power governor model with LFSM-O	DIgSILENT	
gov_PSSE_CRCMGV	Cross Compound Turbine-Governor	PSS/E	Ref: [7], Sec. 7.2
gov_CT1	General model for any prime mover with a PID governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.6
gov_CGMES_3_1_CT1	General model for any prime mover with a PID governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.27
gov_CT2	General governor model with frequency-dependent fuel flow limit	CGMES 2.4.15	Ref [5], Sec. 9.6.6.7
gov_CGMES_3_1_CT2	General governor with frequency-dependent fuel flow limit	CGMES 3.1	Ref: [8], Sec. 8.4.6.28
gov_PSSE_DEGOV	Woodward Diesel Governor	PSS/E	Ref: [7], Sec. 7.3
gov_PSSE_DEGOV1	Woodward Diesel Governor	PSS/E	Ref: [7], Sec. 7.4
gov_PSSE_DEGOV1D	Woodward Diesel Governor with deadband	PSS/E	Ref: [7], Sec. 7.5
gov_GAS	Simplified gas turbine-governor model	DIgSILENT	
gov_IEEE_GAST	Simple gas turbine-governor model	IEEE	Ref: [3]

Type Name	Description	Library	Reference
gov_GAST	Single shaft gas turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.8
gov_CGMES_3_1_GAST	Single shaft gas turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.29
gov_PSSE_GAST	Gas Turbine-Governor	PSS/E	Ref: [7], Sec 7.6.
gov_GAST1	Modified single shaft gas turbine governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.9
gov_CGMES_3_1_GAST1	Modified single shaft gas turbine governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.30
gov_GAST2	Gas turbine model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.10
gov_CGMES_3_1_GAST2	Gas turbine model	CGMES 3.1	Ref: [8], Sec. 8.4.6.31
gov_IEEE_GAST2A	Gas turbine-governor model	IEEE	Ref: [3]
gov_PSSE_GAST2A	Gas Turbine Model	PSS/E	Ref: [7], Sec. 7.7
gov_PSSE_GAST2ADU	Gas Turbine Model	PSS/E	Ref: [7], Sec. 7.8
gov_GAST3	Generic turbo-gas with acceleration and temperature control	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.11
gov_CGMES_3_1_GAST3	Generic Turbo-gas with acceleration and temperature control	CGMES 3.1	Ref: [8], Sec. 8.4.6.32
gov_GAST4	Generic turbo-gas	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.12
gov_CGMES_3_1_GAST4	Generic Turbo-gas	CGMES 3.1	Ref: [8], Sec. 8.4.6.33
gov_PSSE_GASTD	Gas Turbine-Governor with deadband	PSS/E	Ref: [7], Sec. 7.9
gov_GASTWD	Woodward gas turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.13
gov_CGMES_3_1_-GASTWD	Woodward gas turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.34
gov_PSSE_GASTWD	Woodward Gas Turbine-Governor Model	PSS/E	Ref: [7], Sec. 7.10
gov_PSSE_GASTWDD	Woodward Gas Turbine-Governor Model	PSS/E	Ref: [7], Sec. 7.11
gov_GENERAL	Generalised turbine-governor model	DIgSILENT	
gov_IEEE_GGOV1	General purpose gas turbine-governor model	IEEE	Ref: [3], Sec. 3.1.2.3
gov_PSSE_GGOV1	General purpose turbine-governor model	PSS/E	Ref: [7], Sec. 7.12
gov_PSSE_GGOV1D	General purpose turbine-governor model type GGOV1	PSS/E	Ref: [7], Sec. 7.13
gov_IEEE_GT1	Simplified CIGRE gas turbine-governor model	IEEE	Ref: [3], Sec. 3.1.2.5
gov_CGMES_3_1_-IEEEGT1	Gas turbine-governor model type GT1	CGMES 3.1	Ref: [8], Sec. 8.4.6.16 and [3], Sec. 3.1.2.5
gov_HYDRO	Simplified hydro turbine-governor model	DIgSILENT	

Type Name	Description	Library	Reference
gov_Hydro1	Basic hydro turbine governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.14
gov_CGMES_3_1_Hydro1	Basic Hydro turbine governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.35
gov_IEEE_Hydro2	Simple hydro turbine and governor model	IEEE	Ref: [3], Sec. 4.1
gov_Hydro2	IEEE hydro turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.15
gov_CGMES_3_1_Hydro2	Modified IEEE type 2 Hydro turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.36
gov_Hydro3	Modified IEEE hydro turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.16
gov_CGMES_3_1_Hydro3	Modified IEEE Hydro turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.37
gov_Hydro4	Hydro turbine and governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.17
gov_CGMES_3_1_Hydro4X	Hydro turbine and governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.38
gov_HydroDD	Double derivative hydro governor and turbine	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.18
gov_CGMES_3_1_HydroDD	Double derivative hydro governor and turbine	CGMES 3.1	Ref: [8], Sec. 8.4.6.39
gov_HydroFrancis	Hydro Francis unit model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.19
gov_CGMES_3_1_HydroFrancis	Hydro Francis Unit Model	CGMES 3.1	Ref: [8], Sec. 8.4.6.40
gov_HydroIEEE_0	IEEE simplified hydro governor-turbine model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.3
gov_HydroIEEE_2	IEEE hydro turbine-governor model	CGMES 2.4.15	Ref[5], Sec. 9.6.6.4
gov_CGMES_3_1_HydroIEEE0	Simple Hydro Turbine model	CGMES 3.1	Ref: [8], Sec. 8.4.6.24
gov_CGMES_3_1_HydroIEEE2	Modified IEEE type 2 Hydro turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.25
gov_HydroPelton	Hydro Pelton unit model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.20
gov_CGMES_3_1_HydroPelton	Hydro Pelton Unit Model	CGMES 3.1	Ref: [8], Sec. 8.4.6.41
gov_HydroPID	PID governor and turbine	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.21
gov_CGMES_3_1_HydroPID	PID governor and turbine	CGMES 3.1	Ref: [8], Sec. 8.4.6.42
gov_HydroPID2	Hydro turbine and governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.22
gov_CGMES_3_1_HydroPID2	Hydro turbine and governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.43
gov_HydroR	Fourth order lead-lag governor and hydro turbine	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.23
gov_CGMES_3_1_HydroR	Fourth order lead-lag governor and hydro turbine	CGMES 3.1	Ref: [8], Sec. 8.4.6.44

Type Name	Description	Library	Reference
gov_HydroWEH	Woodward electric hydro governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.24
gov_CGMES_3_1_-HydroWEH	Woodward electric hydro governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.45
gov_HydroWPID	Woodward PID hydro governor	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.25
gov_CGMES_3_1_HydroW-PID	Woodward PID hydro governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.46
gov_PSSE_HYG3	Modified IEEE Hydro turbine governor model	PSS/E	Ref: [7], Sec. 7.22
gov_PSSE_HYGOV	Hydro Turbine-Governor	PSS/E	Ref: [7], Sec. 7.15
gov_PSSE_HYGOV2	Hydro Turbine-Governor	PSS/E	Ref: [7], Sec. 7.16
gov_PSSE_HYGOV2D	Hydro Turbine-Governor	PSS/E	Ref: [7], Sec. 7.17
gov_PSSE_HYGOVD	Hydro Turbine-Governor	PSS/E	Ref: [7], Sec. 7.18
gov_PSSE_HYGOVM	Hydro Turbine-Governor Lumped Parameters Model	PSS/E	Ref: [7], Sec. 7.19
gov_PSSE_HYGOVR1	Fourth Order Lead-Lag Hydro Turbine	PSS/E	Ref: [7], Sec. 7.20
gov_IEEE_IEEEG1	Simple steam turbine-governor model	IEEE	Ref: [3], Sec. 2.2
gov_PSSE_IEEEG1	IEEE Type 1 Speed Governing Model	PSS/E	Ref: [7], Sec. 7.24
gov_IEEE_IEEEG1_Augmented	Augmented IEEEG1 steam turbine-governor model with simplified boiler and coordinate control	IEEE	Ref: [3], Sec. 2.3
gov_PSSE_IEEEG1D	Steam turbine governor Type IEEE1 model	PSS/E	Ref: [7], Sec. 7.25
gov_PSSE_IEEEG2	IEEE Type 2 Speed-Governing Model	PSS/E	Ref [7], Sec. 7.26
gov_PSSE_IEEEG3	IEEE Type 3 Speed-Governing Model	PSS/E	Ref [7], Sec. 7.27
gov_PSSE_IEEEG3D	IEEE Type 3 Speed-Governing Model	PSS/E	Ref: [7], Sec. 7.28
gov_IEEE_IEESGO	Simple steam turbine-governor model	IEEE	Ref: [3]
gov_PSSE_IEESGO	IEEE Standard Model	PSS/E	Ref: [7], Sec. 7.29
gov_PSSE_IEESGOD	IEEE Standard Model	PSS/E	Ref: [7], Sec. 7.30
gov_PSSE_IVOGO	IVO Governor Model	PSS/E	Ref: [7], Sec. 7.31
gov_PSSE_PIDGOV	Hydro Turbine-Governor	PSS/E	Ref [7], Sec. 7.32
gov_PSSE_PIDGOVD	Hydro Turbine-Governor	PSS/E	Ref: [7], Sec. 7.33
gov_SIMPLE	Basic turbine-governor model	DIgSILENT	
gov_PSSE_SITGT1	Siemens SGT-A65 Gas Turbine Model	PSS/E	Ref: [7], Sec. 7.38
gov_STEAM	Simplified steam turbine-governor model	DIgSILENT	
gov_Steam0	Simplified steam turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.26

Type Name	Description	Library	Reference
gov_CGMES_3_1_Steam0	Simplified steam turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.47
gov_Steam1	Steam turbine governor model based on the GovSteamIEEE1 model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.27
gov_CGMES_3_1_Steam1	Steam turbine governor model based on the GovSteamIEEE1 model	CGMES 3.1	Ref: [8], Sec. 8.4.6.48
gov_Steam2	Simplified governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.28
gov_CGMES_3_1_Steam2	Simplified governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.49
gov_CGMES_3_1_-SteamBB	Brown Boveri (BB) steam turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.50
gov_SteamCC	Cross compound turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.29
gov_CGMES_3_1_-SteamCC	Cross compound turbine governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.51
gov_SteamEU	Simplified model of boiler and steam turbine	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.30
gov_CGMES_3_1_-SteamEU	Simplified model of boiler and steam turbine	CGMES 3.1	Ref: [8], Sec. 8.4.6.52
gov_PSSE_STEAMEUU	Simplified model of boiler and steam turbine	PSS/E	Ref: [7], Sec. 7.14
gov_SteamFV2	Steam turbine governor with reheat time constants and modelling of the effect of fast valve closing	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.31
gov_CGMES_3_1_-SteamFV2	Steam turbine governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.53
gov_CGMES_3_1_-SteamFV3	Simplified GovSteamIEEE1 steam turbine governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.54
gov_CGMES_3_1_-SteamFV4	Detailed electro-hydraulic governor for steam unit	CGMES 3.1	Ref: [8], Sec. 8.4.6.55
gov_SteamIEEE_1	IEEE steam turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.5
gov_CGMES_3_1_-SteamIEEE1	IEEE Steam turbine-governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.26
gov_CGMES_3_1_-SteamIEEEG1Aug	Simple steam turbine type IEEEG1 with basic boiler dynamics model.	CGMES 3.1	Ref: [8], Sec. 8.4.6.9 and [3], Sec. 2.3
gov_CGMES_3_1_-SteamIEEETGOV5	Detailed steam turbine-governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.8 and [3], Sec. 2.4
gov_SteamSGO	Simplified steam turbine governor model	CGMES 2.4.15	Ref: [5], Sec. 9.6.6.34
gov_CGMES_3_1_-SteamSGO	IEEE Standard Governor	CGMES 3.1	Ref: [8], Sec. 8.4.6.56
gov_IEEE_TGOV1	Simple steam turbine-governor model	IEEE	Ref: [3], Sec. 2.2
gov_PSSE_TGOV1	Steam Turbine-Governor	PSS/E	Ref: [7], Sec. 7.40

Type Name	Description	Library	Reference
gov_PSSE_TGOV1D	Steam Turbine-Governor	PSS/E	Ref: [7], Sec. 7.41
gov_PSSE_TGOV2	Steam Turbine-Governor with Fast Valving	PSS/E	Ref: [7], Sec. 7.42
gov_PSSE_TGOV3	Modified IEEE Type 1 Speed-Governing Model with Fast Valving	PSS/E	Ref: [7], Sec. 7.43
gov_PSSE_TGOV3D	Modified IEEE Type 1 Speed-Governing Model with Fast Valving	PSS/E	Ref: [7], Sec. 7.44
gov_IEEE_TGOV5	Detailed steam turbine-governor model with boiler system	IEEE	Ref: [3], Sec. 2.4
gov_PSSE_TGOV5	IEEE Type 1 Speed-Governing Model Modified to Include Boiler Controls	PSS/E	Ref: [7], Sec. 7.46
gov_PSSE_TURCZT	Czech Hydro and Steam Governor	PSS/E	Ref: [7], Sec. 7.47
gov_PSSE_TWDM1T	Tail Water Depression Hydro Governor Model 1	PSS/E	Ref: [7], 7.48
gov_PSSE_URGS3T	WECC Gas Turbine Model	PSS/E	Ref: [7], Sec. 7.51
gov_PSSE_WEHGOV	WECC Gas Turbine Model	PSS/E	Ref: [7], Sec. 7.52
gov_PSSE_WESGOV	Westinghouse Digital Governor for Gas Turbines	PSS/E	Ref [7], Sec. 7.53
gov_PSSE_WESGOVD	Westinghouse Digital Governor for Gas Turbines	PSS/E	Ref: [7], Sec. 7.54
gov_PSSE_WPIDHY	Woodward PID Hydro Governor	PSS/E	Ref: [7], Sec. 7.55
gov_PSSE_WPIDHYDU	Woodward PID Hydro Governor	PSS/E	Ref: [7], Sec. 7.56
gov_PSSE_WSHYDD	WECC Double Derivative Hydro Governor	PSS/E	Ref: [7], Sec. 7.57
gov_PSSE_WSHYGP	WECC GP Hydro Governor plus Turbine	PSS/E	Ref: [7], Sec. 7.58
gov_PSSE_WSIEG1	WECC Modified IEEE Type 1 Speed-Governing Model	PSS/E	Ref: [7], Sec. 7.59

Table 2.1: Governor and Turbine/Engine Models

2.2 Governor Models

Type Name	Description	Library	Reference
pco_IEEE_Hydro_Mechanical	Mechanical hydraulic governor model	IEEE	Ref: [3], Sec. 4.2
pco_IEEE_HydroDD	Double-derivative hydro governor	IEEE	Ref: [3], Sec. 4.2
pco_CGMES_3_1_IEEEHydroDD	Double-derivative hydro governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.10 and [3], Sec. 4.2
pco_IEEE_HydroLL	Lead-lag hydro governor	IEEE	Ref: [3], Sec. 4.2
pco_CGMES_3_1_IEEEHydroLL	Lead-lag hydro governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.11 and [3], Sec. 4.2

Type Name	Description	Library	Reference
pco_CGMES_3_1_IEEEHydroMech	Mechanical hydraulic governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.12 and [3], Sec. 4.2
pco_IEEE_HydroPID	Hydro PID governor	IEEE	Ref:[3], Sec. 4.2
pco_CGMES_3_1_IEEEHydroPID	PID hydro governor model	CGMES 3.1	Ref: [8], Sec. 8.4.6.13 and [3], Sec. 4.2
pco_IEEE_HydroPID_STPCTRL	Hydro PID governor with setpoint control and nonlinear compensation	IEEE	Ref: [3], Sec. 4.2
pco_CGMES_3_1_IEEEHydroPIDspt	Hydro PID governor with setpoint control and nonlinear compensation model	CGMES 3.1	Ref: [8], Sec. 8.4.6.14 and [3], Sec. 4.2

Table 2.2: Governor Models

Note: Governor models require to be included in a sub-frame “Sub-Frame Turbine and Governor” along with a pmu_* or a trb_* model representing a turbine or engine.

2.3 Prime Mover Models (Turbines, Engines, HRSG)

Type Name	Description	Library	Reference
pmu_CGMES_3_1_CIGREHRSGST	CIGRE heat-recovery steam generator (HRSG) and steam turbine (ST) model	CGMES 3.1	Ref: [8], Sec. 8.4.6.17 and [3], Sec. 3.2.2
pmu_CGMES_3_1_IEEEHRSGST	Generic CCPP heat-recovery steam generator (HRSG) and steam turbine (ST) model	CGMES 3.1	Ref: [8], Sec. 8.4.6.18 and [3], Sec. 3.2.1
pmu_CGMES_3_1_IEEEHydroWCgen	General form of idealized linear turbine-water column model	CGMES 3.1	Ref: [8], Sec. 8.4.6.19 and [3], Sec. 4.3
pmu_CGMES_3_1_IEEEHydroWClinear	Linear turbine-water column model	CGMES 3.1	Ref: [8], Sec. 8.4.6.20 and [3], Sec. 4.3
pmu_CGMES_3_1_IEEEHydroWCNonlinear	Nonlinear turbine-water column model (inelastic)	CGMES 3.1	Ref: [8], Sec. 8.4.6.21 and [3], Sec. 4.3
pmu_CGMES_3_1_IEEEHydroWCNonLinearA	Conventional nonlinear turbine-water column model (inelastic)	CGMES 3.1	Ref: [8], Sec. 8.4.6.22 and [3], Sec. 4.3
pmu_CGMES_3_1_IEEEHydroWCNonLinearB	Modified nonlinear turbine-water column model (inelastic)	CGMES 3.1	Ref: [8], Sec. 8.4.6.23 and [3], Sec. 4.3
trb_IEEE_HRSG_ST_CIGRE	CIGRE multiple-shaft CCPP heat-recovery steam generator (HRSG) and steam turbine (ST)	IEEE	Ref: [3], Sec. 3.2.2
trb_IEEE_HRSG_ST_Generic	Generic CCPP heat-recovery steam generator (HRSG) and steam turbine (ST)	IEEE	Ref: [3], Sec. 3.2.1
trb_IEEE_Hydro_General	General form of idealized linear turbine-water column model	IEEE	Ref: [3], Sec. 4.3
trb_IEEE_Hydro_Linear	Linear turbine-water column model	IEEE	Ref: [3], Sec. 4.3
trb_IEEE_Hydro_Nonlinear	Nonlinear turbine-water column model (inelastic)	IEEE	Ref: [3], Sec. 4.3

Type Name	Description	Library	Reference
trb_IEEE_Hydro_Nonlinear_A	Conventional nonlinear turbine-water column model (inelastic)	IEEE	Ref: [3], Sec. 4.3
trb_IEEE_Hydro_Nonlinear_B	Modified nonlinear turbine-water column model (inelastic)	IEEE	Ref: [3], Sec. 4.3

Table 2.3: Prime Mover Models (Turbines, Engines, HRSG)

Note: Turbine or engine models require to be included in a sub-frame “Sub-Frame Turbine and Governor” along with a pco_* model. HRSG models require to be included in a sub-frame “Sub-Frame CCPP HRSG/ST” . The governor + turbine models in the frame(s) of the gas generator(s) driving the HRSG must be linked in this subframe as well.

2.4 Voltage Transducer and Compensation Models

Type Name	Description	Library	Reference
vcmp_IEEE_1	Voltage compensator IEEE Type 1	CGMES 2.4.15	Ref: [2]
vcmp_IEEE_2	Voltage compensator IEEE Type2	CGMES 2.4.15	Ref: [2]
vtc_IEEE_1	IEEE Type 1 voltage transducer and current compensation	IEEE	Ref: [6], Sec. 5
vtc_CGMES_3_1_IEEE1	IEEE Type 1 voltage transducer and current compensation	CGMES 3.1	Ref: [8], Sec. 8.4.17.3 and [6], Sec. 5
drp_PSSE_COMP	Voltage Regulator Current Compensating Model	PSS/E	Ref: [7], Sec. 2.2
drp_PSSE_REMCMP	Remote measurement interface	PSS/E	Ref: [7], Sec. 2.5
drp_PSSE_IEEEVC	Voltage Regulator Current Compensating Model	PSS/E	Ref: [7], Sec. 2.4

Table 2.4: Voltage Transducer and Compensation Models

2.5 Automatic Voltage Regulator (AVR) and Exciter Models

Type Name	Description	Library	Reference
exc_AC	Simplified AC excitation system model	DIgSILENT	
exc_IEEE_AC10C	IEEE Type AC10C excitation system	IEEE	Ref: [6], Sec. 7.19
exc_CGMES_3_1_-IEEEAC10C	IEEE Type AC10C excitation system	CGMES 3.1	Ref: [8], 8.4.9.40.2 and [6], Sec. 7.19
exc_IEEE_AC11C	IEEE Type AC11C excitation system	IEEE	Ref: [6], Sec. 7.20
exc_CGMES_3_1_-IEEEAC11C	IEEE Type AC11C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.3 and [6], Sec. 7.20

Type Name	Description	Library	Reference
exc_PSSE_AC11C	IEEE Type AC11C excitation system	PSS/E	Ref: [7], Sec. 6.15
exc_IEEE_AC1A	IEEE Type AC1A excitation system	IEEE	Ref: [2], Sec. 6.1
exc_IEEE_AC1A	IEEE Type AC1A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.3
exc_AC1A	Modified IEEE Type AC1A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.22
exc_CGMES_3_1_AC1A	Modified IEEE Type AC1A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.3
exc_CGMES_3_1_-IEEEAC1A	IEEE Type AC1A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.2 and [2], Sec. 6.1
exc_IEEE_AC1C	IEEE Type AC1C excitation system	IEEE	Ref: [6], Sec. 7.3
exc_CGMES_3_1_-IEEEAC1C	IEEE Type AC1C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.4 and [6], Sec. 7.3
exc_PSSE_AC1C	IEEE Type AC1C excitation system	PSS/E	Ref: [7], Sec. 6.1
exc_IEEE_AC2A	IEEE Type AC2A excitation system	IEEE	Ref: [2], Sec. 6.2
exc_IEEE_AC2A	IEEE Type AC2A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_AC2A	Modified IEEE Type AC2A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.23
exc_CGMES_3_1_AC2A	Modified IEEE Type AC2A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.4
exc_CGMES_3_1_-IEEEAC2A	IEEE Type AC2A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.3 and [2], Sec. 6.2
exc_IEEE_AC2C	IEEE Type AC2C excitation system	IEEE	Ref: [6], Sec. 7.5
exc_CGMES_3_1_-IEEEAC2C	IEEE Type AC2C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.5 and [6], Sec. 7.5
exc_PSSE_AC2C	IEEE Type AC2C excitation system	PSS/E	Ref: [7], Sec. 6.2
exc_IEEE_AC3A	IEEE Type AC3A excitation system	IEEE	Ref: [2], Sec. 6.3
exc_IEEE_AC3A	IEEE Type AC3A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_AC3A	Modified IEEE Type AC3A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.24
exc_CGMES_3_1_AC3A	Modified IEEE Type AC3A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.5
exc_CGMES_3_1_-IEEEAC3A	IEEE Type AC3A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.4 and [2], Sec. 6.3
exc_IEEE_AC3C	IEEE Type AC3C excitation system	IEEE	Ref: [6], Sec. 7.7

Type Name	Description	Library	Reference
exc_CGMES_3_1_-IEEEAC3C	IEEE Type AC3C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.6 and [6], Sec. 7.7
exc_PSSE_AC3C	IEEE Type AC3C excitation system	PSS/E	Ref: [7], Sec. 6.3
exc_IEEE_AC4A	IEEE Type AC4A excitation system	IEEE	Ref:[2], Sec. 6.4
exc_IEEE_AC4A	IEEE Type AC4A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_AC4A	Modified IEEE Type AC4A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.25
exc_CGMES_3_1_AC4A	Modified IEEE Type AC4A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.6
exc_CGMES_3_1_-IEEEAC4A	IEEE Type AC4A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.5 and [2], Sec. 6.4
exc_IEEE_AC4C	IEEE Type AC4C excitation system	IEEE	Ref: [6], Sec. 7.9
exc_CGMES_3_1_-IEEEAC4C	IEEE Type AC4C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.7 and [6], Sec. 7.9
exc_PSSE_AC4C	IEEE Type AC4C excitation system	PSS/E	Ref: [7], Sec. 6.4
exc_IEEE_AC5A	IEEE Type AC5A excitation system	IEEE	Ref: [2], Sec. 6.5
exc_IEEE_AC5A	IEEE Type AC5A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_AC5A	Modified IEEE Type AC5A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.26
exc_CGMES_3_1_AC5A	Mod. IEEE AC5A alternator-supplied rectifier exc. system with different minimum controller output	CGMES 3.1	Ref: [8], Sec. 8.4.9.7
exc_CGMES_3_1_-IEEEAC5A	IEEE Type AC5A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.6 and [2], Sec. 6.5
exc_IEEE_AC5C	IEEE Type AC5C excitation system	IEEE	Ref: [6], 7.11
exc_CGMES_3_1_-IEEEAC5C	IEEE Type AC5C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.8 and [6], Sec. 7.11
exc_PSSE_AC5C	IEEE Type AC5C excitation system	PSS/E	Ref: [7], Sec. 6.5
exc_IEEE_AC6A	IEEE Type AC6A excitation system	IEEE	Ref: [2], Sec. 6.6
exc_IEEE_AC6A	IEEE Type AC6A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_AC6A	Modified IEEE Type AC6A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.27
exc_CGMES_3_1_AC6A	Modified IEEE Type AC6A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.8

Type Name	Description	Library	Reference
exc_CGMES_3_1_-IEEEAC6A	IEEE Type AC6A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.7 and [2], Sec. 6.6
exc_PSSE_AC6A	IEEE Type AC6A excitation system	PSS/E	Ref: [7], Sec. 6.6
exc_IEEE_AC6C	IEEE Type AC6C excitation system	IEEE	Ref: [6], Sec. 7.13
exc_CGMES_3_1_-IEEEAC6C	IEEE Type AC6C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.9 and [6], Sec. 7.13
exc_PSSE_AC6C	IEEE Type AC6C excitation system	PSS/E	Ref: [7], Sec. 6.7
exc_PSSE_AC6C2	Modified IEEE Type AC6C excitation system	PSS/E	Ref: [7], Sec. 6.8
exc_IEEE_AC7B	IEEE Type AC7B excitation system	IEEE	Ref: [2], Sec. 6.7
exc_IEEE_AC7B	IEEE Type AC7B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9.9
exc_CGMES_3_1_-IEEEAC7B	IEEE Type AC7B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.8 and [2], Sec. 6.7
exc_PSSE_AC7B	IEEE 421.5-2005 AC7B excitation system	PSS/E	Ref: [7], Sec. 6.9
exc_IEEE_AC7C	IEEE Type AC7C excitation system	IEEE	Ref: [6], Sec. 7.15
exc_CGMES_3_1_-IEEEAC7C	IEEE Type AC7C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.10 and [6], Sec. 7.15
exc_PSSE_AC7C	IEEE Type AC7C excitation system	PSS/E	Ref: [7], Sec. 6.10
exc_IEEE_AC8B	IEEE Type AC8B excitation system	IEEE	Ref: [2], Sec. 6.8
exc_IEEE_AC8B	IEEE Type AC8B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9.10
exc_AC8B	Modified IEEE Type AC8B excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.28
exc_CGMES_3_1_AC8B	Modified IEEE Type AC8B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.9
exc_CGMES_3_1_-IEEEAC8B	IEEE Type AC8B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.9 and [2], Sec. 6.8
exc_PSSE_AC8B	IEEE 421.5-2005 AC8B excitation system	PSS/E	Ref: [7], Sec. 6.11
exc_IEEE_AC8C	IEEE Type AC8C excitation system	IEEE	Ref: [6], Sec. 7.17
exc_CGMES_3_1_-IEEEAC8C	IEEE Type AC8C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.11 and [6], v7.17
exc_PSSE_AC8C	IEEE Type AC8C excitation system	PSS/E	Ref: [7], Sec. 6.13

Type Name	Description	Library	Reference
exc_IEEE_AC9C	IEEE Type AC9C excitation system	IEEE	Ref: [6], Sec. 7.18
exc_CGMES_3_1_-IEEEAC9C	IEEE Type AC9C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.12 and [6], Sec. 7.18
exc_PSSE_AC9C	IEEE Type AC9C excitation system	PSS/E	Ref: [7], Sec. 6.14
exc_ANS	Italian excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.29
exc_CGMES_3_1_ANS	Italian excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.10
exc_AVR1	Italian excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.30
exc_CGMES_3_1_AVR1	IEEE (1968) Type1 excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.11
exc_CGMES_3_1_AVR2	IEEE (1968) Type2 excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.12
exc_AVR2	Italian excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.31
exc_AVR3	Italian excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.32
exc_CGMES_3_1_AVR3	Italian excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.13
exc_AVR4	Italian excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.33
exc_CGMES_3_1_AVR4	Italian excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.14
exc_AVR5	Manual excitation control with field circuit resistance	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.34
exc_CGMES_3_1_AVR5	Manual excitation control with field circuit resistance	CGMES 3.1	Ref: [8], Sec. 8.4.9.15
exc_AVR7	IVO excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.35
exc_CGMES_3_1_AVR7	IVO excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.16
exc_BBC	Transformer fed static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.36
exc_CGMES_3_1_BBC	Transformer fed static excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.17
exc_PSSE_BBSEX1	Brown Boveri Static Exciter	PSS/E	Ref: [7], Sec. 6.16
exc_PSSE_BUDCZT	Czech Proportional/Integral Exciter	PSS/E	Ref: [7], Sec. 6.17
exc_PSSE_CELIN	ELIN Excitation System	PSS/E	Ref: [7], Sec. 6.18
exc_CZ	Czech proportion/integral exciter	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.37
exc_CGMES_3_1_CZ	Czech proportion/integral exciter	CGMES 3.1	Ref: [8], Sec. 8.4.9.18
exc_IEEE_DC1A	IEEE Type DC1A excitation system	IEEE	Ref: [2], Sec. 5.1

Type Name	Description	Library	Reference
exc_IEEE_DC1A	IEEE Type DC1A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_DC1A	Modified IEEE Type DC1A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.38
exc_CGMES_3_1_DC1A	Modified IEEE Type DC1A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.19
exc_CGMES_3_1_-IEEEDC1A	IEEE Type DC1A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.10 and [2], Sec. 5.1
exc_IEEE_DC1C	IEEE Type DC1C excitation system	IEEE	Ref: [6], Sec. 6.3
exc_CGMES_3_1_-IEEEDC1C	IEEE Type DC1C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.13 and [6], Sec. 6.3
exc_PSSE_DC1C	IEEE Type DC1C excitation system	PSS/E	Ref: [7], Sec. 6.19
exc_IEEE_DC2A	IEEE Type DC2A excitation system	IEEE	Ref: [2], Sec. 5.2
exc_IEEE_DC2A	IEEE Type DC2A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9.12
exc_DC2A	Modified IEEE Type DC2A excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.39
exc_CGMES_3_1_DC2A	Modified IEEE Type DC2A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.20
exc_CGMES_3_1_-IEEEDC2A	IEEE Type DC2A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.11 and [2], Sec. 5.2
exc_IEEE_DC2C	IEEE Type DC2C excitation system	IEEE	Ref: [6], 6.5
exc_CGMES_3_1_-IEEEDC2C	IEEE Type DC2C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.14 and [6], Sec. 6.5
exc_PSSE_DC2C	IEEE Type DC2C excitation system	PSS/E	Ref: [7], Sec. 6.20
exc_IEEE_DC3A	IEEE Type DC3A excitation system	IEEE	Ref: [2], Sec. 5.3
exc_IEEE_DC3A	IEEE Type DC3A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9.13
exc_DC3A	Modified IEEE DC3A direct current commutator exciter	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.40
exc_CGMES_3_1_DC3A	Modified IEEE DC3A direct current commutator exciter	CGMES 3.1	Ref: [8], Sec. 8.4.9.21
exc_CGMES_3_1_-IEEEDC3A	IEEE Type DC3A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.12 and [2], Sec. 5.3
exc_PSSE_DC3A	IEEE 421-5-2005 DC3A Excitation System	PSS/E	Ref: [7], Sec. 6.21
exc_DC3A1	Modified IEEE Type 3 excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.41
exc_CGMES_3_1_DC3A1	Modified IEEE Type 3 excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.22

Type Name	Description	Library	Reference
exc_IEEE_DC4B	IEEE Type DC4B excitation system	IEEE	Ref: [2], Sec. 5.4
exc_IEEE_DC4B	IEEE Type DC4B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_-IEEEDC4B	IEEE Type DC4B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.13 and [2], Sec. 5.4
exc_PSSE_DC4B	IEEE 421.5-2005 DC4B Excitation System	PSS/E	Ref: [7], Sec. 6.22
exc_IEEE_DC4C	IEEE Type DC4C excitation system	IEEE	Ref: [6], Sec. 6.8
exc_CGMES_3_1_-IEEEDC4C	IEEE Type DC4C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.15 and [6], Sec. 6.8
exc_PSSE_DC4C	IEEE Type DC4C excitation system	PSS/E	Ref: [7], Sec. 6.23
exc_ELIN1	Static PI transformer fed excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.42
exc_CGMES_3_1_ELIN1	Static PI transformer fed excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.23
exc_ELIN2	ELIN (VATECH) detailed excitation system model	CGMES 2.4.15	Ref: [5], 9.6.9.43
exc_CGMES_3_1_ELIN2	ELIN (VATECH) detailed excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.24
exc_PSSE_EMAC1T	Modified IEEE Type AC1 Excitation System	PSS/E	Ref: [7], Sec. 6.24
exc_PSSE_ESAC1A	IEEE Type AC1A excitation system	PSS/E	Ref: [7], Sec. 6.25
exc_PSSE_ESAC2A	IEEE Type AC2A excitation system	PSS/E	Ref: [7], Sec. 6.26
exc_PSSE_ESAC3A	IEEE Type AC3A excitation system	PSS/E	Ref: [7], Sec. 6.27
exc_PSSE_ESAC4A	IEEE Type AC4A excitation system	PSS/E	Ref: [7], Sec 6.28
exc_PSSE_ESAC5A	IEEE Type AC5A Excitation System	PSS/E	Ref: [7], Sec 6.29
exc_PSSE_ESAC6A	IEEE Type AC6A excitation system	PSS/E	Ref: [7], Sec 6.30
exc_PSSE_ESAC8B	Basler DECS	PSS/E	Ref: [7], Sec. 6.31
exc_PSSE_ESDC1A	IEEE Type DC1A Excitation System	PSS/E	Ref: [7], Sec. 6.32
exc_PSSE_ESDC2A	IEEE Type DC2A Excitation System	PSS/E	Ref: [7], Sec. 6.33
exc_PSSE_ESST1A	IEEE Type ST1A Excitation System	PSS/E	Ref: [7], Sec. 6.34
exc_PSSE_ESST2A	Modified IEEE Type ST2A Excitation System	PSS/E	Ref: [7], Sec. 6.35
exc_PSSE_ESST3A	IEEE Type ST3A Excitation System	PSS/E	Ref: [7], Sec. 6.36

Type Name	Description	Library	Reference
exc_PSSE_ESST4B	IEEE Type ST4B Potential or Compounded Source-Controlled Rectifier Exciter	PSS/E	Ref: [7], Sec. 6.37
exc_PSSE_ESURRY	Modified IEEE Type AC1A Excitation System	PSS/E	Ref: [7], Sec. 6.38
exc_PSSE_EX2000	EX2000 Excitation System	PSS/E	Ref: [7], Sec. 6.39
exc_PSSE_EXAC1	IEEE Type AC1 Excitation System	PSS/E	Ref: [7], Sec. 6.40
exc_PSSE_EXAC1A	Modified IEEE Type AC1 Excitation System	PSS/E	Ref: [7], Sec. 6.41
exc_PSSE_EXAC2	IEEE Type AC2 Excitation System	PSS/E	Ref: [7], Sec. 6.42
exc_PSSE_EXAC3	IEEE Type AC3 Excitation System	PSS/E	Ref: [7], Sec. 6.43
exc_PSSE_EXAC4	1981 IEEE Type AC4 Excitation System	PSS/E	Ref: [7], Sec. 6.44
exc_PSSE_EXBAS	Basler Static Voltage Regulator Feeding DC or AC Rotating Exciter	PSS/E	Ref: [7], Sec. 6.45
exc_PSSE_EXDC2	IEEE Type DC2 Excitation System	PSS/E	Ref: [7], Sec. 6.46
exc_PSSE_EXELI	Static PI Transformer Fed Excitation System	PSS/E	Ref: [7], Sec. 6.47
exc_PSSE_EXNEBB	Bus or Solid Fed SCR Bridge Excitation System Model	PSS/E	Ref: [7], Sec. 6.48
exc_PSSE_EXNI	Bus or Solid Fed SCR Bridge Excitation System Model	PSS/E	Ref: [7], Sec. 6.49
exc_PSSE_EXPIC1	Proportional/Integral Excitation System	PSS/E	Ref: [7], Sec. 6.50
exc_PSSE_EXST1	IEEE Type ST1 Excitation System	PSS/E	Ref: [7], Sec. 6.51
exc_PSSE_EXST2	IEEE Type ST2 Excitation System	PSS/E	Ref: [7], Sec. 6.52
exc_PSSE_EXST2A	Modified IEEE Type ST2 Excitation System	PSS/E	Ref: [7], Sec. 6.53
exc_PSSE_EXST3	IEEE Type ST3 Excitation System	PSS/E	Ref: [7], Sec. 6.54
exc_HU	Hungarian excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.44
exc_CGMES_3_1_HU	Hungarian excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.25
exc_PSSE_IEEET1	IEEE Type 1 Excitation System	PSS/E	Ref: [7], Sec. 6.55
exc_PSSE_IEEET2	IEEE Type 2 Excitation System	PSS/E	Ref: [7], Sec. 6.56
exc_PSSE_IEEET3	IEEE Type 3 Excitation System	PSS/E	Ref: [7], Sec. 6.57
exc_PSSE_IEEET4	IEEE Type 4 Excitation System	PSS/E	Ref: [7], Sec. 6.58
exc_PSSE_IEEET5	Modified IEEE Type 4 Excitation System	PSS/E	Ref: [7], Sec. 6.59
exc_PSSE_IEEEX1	IEEE Type 1 Excitation System	PSS/E	Ref: [7], Sec. 6.60
exc_PSSE_IEEEX2	IEEE Type 2 Excitation System	PSS/E	Ref: [7], Sec. 6.61

Type Name	Description	Library	Reference
exc_PSSE_IEEEX3	IEEE Type 3 Excitation System	PSS/E	Ref: [7], Sec. 6.62
exc_PSSE_IEEEX4	IEEE Type 4 Excitation System	PSS/E	Ref: [7], Sec. 6.63
exc_PSSE_IEET1A	Modified IEEE Type 1 Excitation System	PSS/E	Ref: [7], Sec. 6.64
exc_PSSE_IEET1B	Modified IEEE Type 1 Excitation System	PSS/E	Ref: [7], Sec. 6.65
exc_PSSE_IEET5A	Modified IEEE Type 4 Excitation System	PSS/E	Ref: [7], Sec. 6.66
exc_PSSE_IEEX2A	IEEE Type 2A Excitation System	PSS/E	Ref: [7], Sec. 6.67
exc_PSSE_IVOEX	IVO Excitation Model	PSS/E	Ref: [7], Sec. 6.68
exc_CGMES_3_1_NI	Bus or solid fed SCR bridge excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.26
exc_PSSE_OEX12T	Ontario Hydro IEEE Type ST1 Excitation System	PSS/E	Ref: [7], Sec. 6.69
exc_OEX3T	Modified IEEE Type ST1 excitation system with semi-continuous and acting terminal voltage limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.45
exc_CGMES_3_1_OEX3T	Modified IEEE Type ST1 excitation system with semi-continuous and acting terminal voltage limiter	CGMES 3.1	Ref: [8], Sec. 8.4.9.27
exc_PSSE_OEX3T	Ontario Hydro IEEE Type ST1 Excitation System with Semi-continuous and Acting Terminal Voltage Limiter	PSS/E	Ref: [7], Sec. 6.70
exc_PIC	Proportional/integral regulator excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.46
exc_CGMES_3_1_PIC	Proportional/Integral Regulator Excitation System Model	CGMES 3.1	Ref: [8], Sec. 8.4.9.28
exc_REXS	General purpose rotating excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.47
exc_CGMES_3_1_REXS	General purpose rotating excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.29
exc_PSSE_REXSY1	General Purpose Rotating Excitation System Model	PSS/E	Ref: [7], Sec. 6.71
exc_PSSE_REXSYS	General Purpose Rotating Excitation System Model	PSS/E	Ref: [7], Sec. 6.72
exc_CGMES_3_1_RQB	Excitation system (four-loop regulator)	CGMES 3.1	Ref: [8], Sec. 8.4.9.30
exc_SCRX	Simple excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.48
exc_CGMES_3_1_SCRX	Simple excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.31
exc_PSSE_SCRX	Bus Fed or Solid Fed Static Exciter	PSS/E	Ref: [7], Sec. 6.73
exc_SEXS	Simplified excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.49
exc_CGMES_3_1_SEXS	Simplified excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.32

Type Name	Description	Library	Reference
exc_PSSE_SEXS	Simplified Excitation System	PSS/E	Ref: [7], Sec. 6.74
exc_SIMPLE	Basic excitation system model	DIgSILENT	
exc_SK	Slovakian excitation system model	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.50
exc_CGMES_3_1_SK	Slovakian excitation system model	CGMES 3.1	Ref: [8], Sec. 8.4.9.33
exc_ST	Simplified static excitation system model	DIgSILENT	
exc_IEEE_ST10C	IEEE Type ST10C excitation system	IEEE	Ref: [6], Sec. 8.18
exc_CGMES_3_1_-IEEEST10C	IEEE Type ST10C excitation system	CGMES 3.1	Ref: [8], 8.4.9.40.16 and [6], 8.18
exc_PSSE_ST10C	IEEE Type ST10C excitation system	PSS/E	Ref: [7], Sec. 6.85
exc_ST1A	Modified IEEE ST1A static excitation system without EOL and UEL	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.51
exc_CGMES_3_1_ST1A	Modified IEEE ST1A static excitation system without EOL and UEL	CGMES 3.1	Ref: [8], Sec. 8.4.9.34
exc_IEEE_ST1A	IEEE Type ST1A excitation system	IEEE	Ref: [2], Sec. 7.1
exc_IEEE_ST1A	IEEE Type ST1A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_-IEEEST1A	IEEE Type ST1A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.14 and [2], Sec. 7.1
exc_IEEE_ST1C	IEEE Type ST1C excitation system	IEEE	Ref: [6], 8.3
exc_CGMES_3_1_-IEEEST1C	IEEE Type ST1C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.17 and [6], Sec. 8.3
exc_PSSE_ST1C	IEEE Type ST1C excitation system	PSS/E	Ref: [7], Sec. 6.75
exc_ST2A	Modified IEEE ST2A static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.52
exc_IEEE_ST2A	IEEE Type ST2A excitation system	IEEE	Ref: [2], Sec. 7.2
exc_IEEE_ST2A	IEEE Type ST2A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_ST2A	Modified IEEE ST2A static excitation system	CGMES 3.1	Ref: [8], 8.4.9.35
exc_CGMES_3_1_-IEEEST2A	IEEE Type ST2A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.15 and [2], Sec. 7.2
exc_IEEE_ST2C	IEEE Type ST2C excitation system	IEEE	Ref: [6], 8.5
exc_CGMES_3_1_-IEEEST2C	IEEE Type ST2C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.18 and [6], Sec. 8.5

Type Name	Description	Library	Reference
exc_ST3A	Modified IEEE ST3A static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.53
exc_IEEE_ST3A	IEEE Type ST3A excitation system	IEEE	Ref: [2], Sec. 7.3
exc_IEEE_ST3A	IEEE Type ST3A excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_ST3A	Modified IEEE ST3A static excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.36
exc_CGMES_3_1_-IEEEST3A	IEEE Type ST3A excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.16 and [2], Sec. 7.3
exc_IEEE_ST3C	IEEE Type ST3C excitation system	IEEE	Ref: [6], Sec. 8.7
exc_CGMES_3_1_-IEEEST3C	IEEE Type ST3C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.19 and [6], Sec. 8.7
exc_ST4B	Modified IEEE ST4B static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.54
exc_IEEE_ST4B	IEEE Type ST4B excitation system	IEEE	Ref: [2], Sec. 7.4
exc_IEEE_ST4B	IEEE Type ST4B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_ST4B	Modified IEEE ST4B static excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.37
exc_CGMES_3_1_-IEEEST4B	IEEE Type ST4B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.17 and [2], Sec. 7.4
exc_IEEE_ST4C	IEEE Type ST4C excitation system	IEEE	Ref: [6], Sec. 8.9
exc_CGMES_3_1_-IEEEST4C	IEEE Type ST4C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.20 and [6], Sec. 8.9
exc_PSSE_ST4C	IEEE Type ST4C excitation system	PSS/E	Ref: [7], Sec. 6.76
exc_IEEE_ST5B	IEEE Type ST5B excitation system	IEEE	Ref: [2], Sec. 7.5
exc_IEEE_ST5B	IEEE Type ST5B excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.19
exc_CGMES_3_1_-IEEEST5B	IEEE Type ST5B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.18 and [2], Sec. 7.5
exc_PSSE_ST5B	IEEE 421.5-2005 ST5B Excitation System	PSS/E	Ref: [7], Sec. 6.77
exc_IEEE_ST5C	IEEE Type ST5C excitation system	IEEE	Ref: [6], Sec. 8.11
exc_CGMES_3_1_-IEEEST5C	IEEE Type ST5C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.21 and [6], Sec. 8.11
exc_PSSE_ST5C	IEEE Type ST5C excitation system	PSS/E	Ref: [7], Sec. 6.78

Type Name	Description	Library	Reference
exc_ST6B	Modified IEEE ST6B static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.55
exc_IEEE_ST6B	IEEE Type ST6B excitation system	IEEE	Ref: [2], Sec. 7.6
exc_IEEE_ST6B	IEEE Type ST6B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_ST6B	Modified IEEE ST6B static excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.38
exc_CGMES_3_1_-IEEEST6B	IEEE Type ST6B excitation system	CGMES 3.1	Ref: [8], 8.4.9.41.19 and [2], Sec. 7.6
exc_PSSE_ST6B	IEEE 421.5-2005 ST6B Excitation System	PSS/E	Ref: [7], Sec. 6.79
exc_IEEE_ST6C	IEEE Type ST6C excitation system	IEEE	Ref: [6], Sec. 8.13
exc_CGMES_3_1_-IEEEST6C	IEEE Type ST6C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.22 and [6], Sec. 8.13
exc_PSSE_ST6C	IEEE Type ST6C excitation system	PSS/E	Ref: [7], Sec. 6.80
exc_ST7B	Modified IEEE ST7B static excitation system	CGMES 2.4.15	Ref: [5], Sec. 9.6.9.56
exc_IEEE_ST7B	IEEE Type ST7B excitation system	IEEE	Ref: [2], Sec. 7.7
exc_IEEE_ST7B	IEEE Type ST7B excitation system	CGMES 2.4.15	Ref:[5], Sec. 9.6.9
exc_CGMES_3_1_ST7B	Modified IEEE ST7B static excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.39
exc_CGMES_3_1_-IEEEST7B	IEEE Type ST7B excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.41.20 and [2], Sec. 7.7
exc_PSSE_ST7B	IEEE 421.5-2005 ST7B Excitation System	PSS/E	Ref: [7], Sec. 6.81
exc_IEEE_ST7C	IEEE Type ST7C excitation system	IEEE	Ref: [6], Sec. 8.15
exc_CGMES_3_1_-IEEEST7C	IEEE Type ST7C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.23 and [6], Sec. 8.15
exc_PSSE_ST7C	IEEE Type ST7C excitation system	PSS/E	Ref: [7], Sec. 6.82
exc_IEEE_ST8C	IEEE Type ST8C excitation system	IEEE	Ref: [6], Sec. 8.16
exc_CGMES_3_1_-IEEEST8C	IEEE Type ST8C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.24 and [6], Sec. 8.16
exc_PSSE_ST8C	IEEE Type ST8C excitation system	PSS/E	Ref: [7], Sec. 6.83
exc_IEEE_ST9C	IEEE Type ST9C excitation system	IEEE	Ref: [6], Sec. 8.17
exc_CGMES_3_1_-IEEEST9C	IEEE Type ST9C excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.9.40.25 and [6], Sec. 8.17

Type Name	Description	Library	Reference
exc_PSSE_ST9C	IEEE Type ST9C excitation system	PSS/E	Ref: [7], Sec. 6.84
exc_PSSE_URHIDT	High Dam Excitation Model	PSS/E	Ref: [7], Sec. 6.86
exc_PSSE_URST5T	IEEE Proposed Type ST5B Excitation System	PSS/E	Ref: [7], Sec. 6.87
exc_PSSE_USAC6AU	Modified ESAC6A Model	PSS/E	Ref: [4], Sec. 6.62

Table 2.5: Excitation System Models

2.6 Power System Stabilizer Models

Type Name	Description	Library	Reference
pss_PSSE_BEPSST	Transient Excitation Boosting PSS	PSS/E	Ref: [7], Sec. 3.1
pss_ELIN2	Power system stabilizer	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.12
pss_CGMES_3_1_ELIN2	Power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.18
pss_PSSE_IEE2ST	IEEE Stabilizing Model with Dual Input Signals	PSS/E	Ref: [7], Sec. 3.2
pss_PSSE_IVOST	IVO Stabilizer Model	PSS/E	Ref: [7], Sec. 3.4
pss_PSSE_OSTB2T	Ontario Hydro Delta-Omega Power System Stabilizer	PSS/E	Ref: [7], Sec. 3.5
pss_PSS1	Italian PSS - Three inputs power system stabilizer	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.7
pss_CGMES_3_1_PSS1	Italian PSS - Three inputs power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.13
pss_IEEE_PSS1A	IEEE Type PSS1A power system stabilizer	IEEE	Ref: [2], Sec. 8.1
pss_PSS1A	Single input power system stabilizer	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.8
pss_CGMES_3_1_PSS1A	Single input power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.14
pss_IEEE_1A	IEEE Type PSS1A - Single-input power system stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.3
pss_IEEE_PSS2A	IEEE Type PSS2A power system stabilizer	IEEE	Ref: [1]
pss_PSSE_PSS2A	IEEE Dual Input Stabilizer Model	PSS/E	Ref: [7], Sec. 3.8
pss_IEEE_PSS2B	IEEE Type PSS2B power system stabilizer	IEEE	Ref: [2], Sec. 8.2
pss_PSS2B	Modified IEEE Type PSS2B power system stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.9
pss_CGMES_3_1_PSS2B	Modified IEEE Type PSS2B - Dual-input power system stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.15
pss_PSSE_PSS2B	IEEE 421.5-2005 PSS2B Dual-Input Stabilizer Model	PSS/E	Ref: [7], Sec. 3.9

Type Name	Description	Library	Reference
pss_IEEE_2B	IEEE Type PSS2B - Dual-input power system stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.4
pss_IEEE_PSS2C	IEEE Type PSS2C power system stabilizer	IEEE	Ref:[6], Sec. 9.5
pss_PSSE_PSS2C	IEEE Type PSS2C power system stabilizer	PSS/E	Ref: [7], Sec. 3.10
pss_PSS2ST	PTI microprocessor-based power system stabilizer type 1	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.10
pss_CGMES_3_1_PSS2ST	PTI microprocessor-based stabilizer type 1	CGMES 3.1	Ref: [8], Sec. 8.4.12.16
pss_IEEE_PSS3B	IEEE Type PSS3B power system stabilizer	IEEE	Ref: [2], Sec. 8.3
pss_PSSE_PSS3B	IEEE Std. 421.5-2005 PSS3B Dual Input Stabilizer Model	PSS/E	Ref: [7], Sec. 3.11
pss_IEEE_3B	IEEE Type PSS3B - Dual-input power system stabiliser model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.5
pss_IEEE_PSS3C	IEEE Type PSS3C power system stabilizer	IEEE	Ref:[6], Sec. 9.7
pss_IEEE_PSS4B	IEEE Type PSS4B power system stabilizer	PSS/E	Ref: [2], Sec. 8.4
pss_PSSE_PSS4B	IEEE 421.5(2005) Dual-Input Stabilizer Model	PSS/E	Ref: [7], Sec. 3.12
pss_IEEE_4B	IEEE Type PSS4B - Three bands power system stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.6
pss_IEEE_PSS4C	IEEE Type PSS4C power system stabilizer	IEEE	Ref: [6], Sec. 9.9
pss_PSS5	Italian PSS - Detailed power system stabilizer model	CGMES 2.4.15	Ref: [5], 9.6.12.11
pss_CGMES_3_1_PSS5	Italian PSS - Detailed power system stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.17
pss_IEEE_PSS5C	IEEE Type PSS5C power system stabilizer	IEEE	Ref:[6], Sec. 9.10
pss_IEEE_PSS6C	IEEE Type PSS6C power system stabilizer	IEEE	Ref: [6], Sec. 9.11
pss_PSSE_PSS6C	IEEE Type PSS6C power system stabilizer	PSS/E	Ref: [7], Sec. 3.13
pss_IEEE_PSS7C	IEEE Type PSS7C power system stabilizer	IEEE	Ref: [6], Sec. 9.12
pss_PSSE_PSS7C	IEEE Type PSS7C power system stabilizer	PSS/E	Ref: [7], Sec. 3.14
pss_CGMES_3_1_-PssIEEE1A	IEEE Type PSS1A - Single-Input Stabilizer Model	CGMES 3.1	Ref: [8], Sec. 8.4.12.9 and [2], Sec. 8.1
pss_CGMES_3_1_-PssIEEE2B	IEEE Type PSS2B - Dual-input power system stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.10 and [2], Sec. 8.2
pss_CGMES_3_1_-PssIEEE2C	IEEE Type PSS2C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.3 and [6], Sec. 9.5

Type Name	Description	Library	Reference
pss_CGMES_3_1_-PssIEEE3B	IEEE Type PSS3B - Dual-input power system stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.11 and [2], Sec. 8.3
pss_CGMES_3_1_-PssIEEE3C	IEEE Type PSS3C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.4 and [6], Sec. 9.7
pss_CGMES_3_1_-PssIEEE4B	IEEE Type PSS4B power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.12 and [2], Sec. 8.4
pss_CGMES_3_1_-PssIEEE4C	IEEE Type PSS4C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.5 and [6], Sec. 9.9
pss_CGMES_3_1_-PssIEEE5C	IEEE Type PSS5C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.6 and [6], Sec. 9.10
pss_CGMES_3_1_-PssIEEE6C	IEEE Type PSS6C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.7 and [6], Sec. 9.11
pss_CGMES_3_1_-PssIEEE7C	IEEE Type PSS7C power system stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.8 and [6], Sec. 9.12
pss_PSSSB4	Power sensitive stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.15
pss_PSSSH	Siemens “H infinity” power system stabilizer model	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.16
pss_PSSSK	Slovakian power system stabilizer type with three inputs	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.17
pss_CGMES_3_1_PTIST1	PTI Micro-Processor Based Stabilizer	CGMES 3.1	Ref: [8], Sec. 8.4.12.19
pss_PSSE_PTIST1	PTI Micro-Processor Based Stabilizer	PSS/E	Ref: [7], Sec. 3.15
pss_CGMES_3_1_RQB	Power system stabilizer type RQB	CGMES 3.1	Ref: [8], Sec. 8.4.12.21
pss_CGMES_3_1_SB4	Power sensitive stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.22
pss_CGMES_3_1_SH	Siemens “H infinity” power system stabilizer model	CGMES 3.1	Ref: [8], Sec. 8.4.12.23
pss_CGMES_3_1_SK	Slovakian power system stabilizer type with three inputs	CGMES 3.1	Ref: [8], Sec. 8.4.12.24
pss_PSSE_IEEEST	IEEE Stabilizing Model	PSS/E	Ref: [7], Sec. 3.3
pss_PSSE_ST2CUT	Stabilizing Model with Dual-Input Signals	PSS/E	Ref: [7], Sec. 3.17
pss_PSSE_STAB1	Speed Sensitive Stabilizing Model	PSS/E	Ref: [7], Sec. 3.18
pss_CGMES_3_1_STAB2A	Power system stabilizer part of an ABB excitation system	CGMES 3.1	Ref: [8], Sec. 8.4.12.25
pss_PSSE_STAB2A	Power Sensitive Stabilizing Unit (ASEA)	PSS/E	Ref: [7], 3.19
pss_PSSE_STAB3	Power Sensitive Stabilizing Unit	PSS/E	Ref: [7], Sec. 3.20
pss_PSSE_STAB4	Power Sensitive Stabilizer	PSS/E	Ref: [7], Sec. 3.21
pss_PSSE_STABNI	Power Sensitive Stabilizer Model	PSS/E	Ref: [7], Sec. 3.22
pss_WECC	Power system stabilizer	CGMES 2.4.15	Ref: [5], Sec. 9.6.12.18

Type Name	Description	Library	Reference
pss_CGMES_3_1_WECC	Dual input PSS, based on IEEE type 2, with modified output limiter defined by WECC	CGMES 3.1	Ref: [8], Sec. 8.4.12.26

Table 2.6: Power System Stabilizer Models

2.7 Discontinuous Excitation Control (DEC) Models

Type Name	Description	Library	Reference
dec_CGMES_3_1_-IEEEDEC1A	IEEE Type DEC1A discontinuous excitation control	CGMES 3.1	Ref: [8], Sec. 8.4.13.3
dec_IEEE_DEC1A	IEEE Type DEC1A discontinuous excitation control	IEEE	Ref: [6], Sec. 14.2
dec_CGMES_3_1_-IEEEDEC3A	IEEE Type DEC3A discontinuous excitation control	CGMES 3.1	Ref: [8], Sec. 8.4.13.5
dec_IEEE_DEC3A	IEEE Type DEC3A discontinuous excitation control	IEEE	Ref: [6], Sec. 14.4

Table 2.7: Discontinuous Excitation Control Models

Note: DEC models require to be included in sub-frame “Sub-Frame PSS/DEC” along with the pss_* model.

2.8 Underexcitation Limiter Models

Type Name	Description	Library	Reference
uel_IEEE1	Underexcitation limiter Type UEL1	CGMES 2.4.15	Ref: [5], Sec. 9.6.11.3
uel_IEEE2	Underexcitation limiter Type UEL2	CGMES 2.4.15	Ref: [5], Sec. 9.6.11.4
uel_2Simplified	Simplified underexcitation limiter (UEL) of type 2	CGMES 2.4.15	Ref: [5], Sec. 9.6.11.5
uel_PSSE_MNLEX1	Minimum Excitation Limiter	PSS/E	Ref: [7], Sec. 4.1
uel_PSSE_MNLEX2	Minimum Excitation Limiter	PSS/E	Ref: [7], Sec. 4.2
uel_PSSE_MNLEX3	Minimum Excitation Limiter	PSS/E	Ref: [7], Sec. 4.3
uel_IEEE_UEL1	IEEE Type UEL1 underexcitation limiter	IEEE	Ref: [6], Sec. 11.2
uel_PSSE_UEL1	IEEE 421.5-2005 UEL1 Under-Excitation Limiter	PSS/E	Ref: [7], Sec. 4.4
uel_IEEE_UEL2	IEEE Type UEL2 underexcitation limiter	IEEE	Ref: [6], Sec. 11.3
uel_PSSE_UEL2	IEEE 421.5-2005 UEL2 Under-Excitation Limiter	PSS/E	Ref: [7], Sec. 4.5
uel_IEEE_UEL2C	IEEE Type UEL2C underexcitation limiter	IEEE	Ref: [6], Sec. 11.4

Type Name	Description	Library	Reference
uel_PSSE_UEL2C	IEEE Type UEL2C underexcitation limiter	PSS/E	Ref: [7], Sec. 4.6
uel_CGMES_3_1_UnderexcLim2S	Simplified type UEL2 underexcitation limiter	CGMES 3.1	Ref: [8], Sec. 8.4.11.6
uel_CGMES_3_1_UnderexcLimIEEE1	Underexcitation Limiter Type UEL1	CGMES 3.1	Ref: [8], Sec. 8.4.11.4 and [6], Sec. 11.2
uel_CGMES_3_1_UnderexcLimIEEE2	Underexcitation Limiter Type UEL2	CGMES 3.1	Ref: [8], Sec. 8.4.11.5 and [6], Sec. 11.3
uel_CGMES_3_1_UnderexcLimIEEEUEL2C	IEEE Type UEL2C underexcitation limiter	CGMES 3.1	Ref: [8], Sec. 8.4.11.2 and [6], Sec. 11.4
uel_CGMES_3_1_UnderexcLimX1	Allis-Chalmers Under Excitation Limiter	CGMES 3.1	Ref: [8], Sec. 8.4.11.7
uel_CGMES_3_1_UnderexcLimX2	Westinghouse Under Excitation Limiter	CGMES 3.1	Ref: [8], Sec. 8.4.11.8
uel_X1	Allis-Chalmers minimum excitation limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.11.6
uel_X2	Westinghouse minimum excitation limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.11.7

Table 2.8: Underexcitation Limiter Models

2.9 Overexcitation Limiter Models

Type Name	Description	Library	Reference
oel_PSSE_MAXEX1	Maximum Excitation Limiter	PSS/E	Ref: [7], Sec. 5.2
oel_PSSE_MAXEX2	Maximum Excitation Limiter	PSS/E	Ref: [7], Sec. 5.3
oel_OEL2	Overexcitation limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.10.4
oel_IEEE_OEL2C	IEEE Type OEL2C overexcitation limiter	IEEE	Ref: [6], Sec. 10.5
oel_PSSE_OEL2C	IEEE Type OEL2C overexcitation limiter	PSS/E	Ref: [7], Sec. 5.4
oel_IEEE_OEL3C	IEEE Type OEL3C overexcitation limiter	IEEE	Ref: [6], Sec. 10.6
oel_IEEE_OEL4C	IEEE Type OEL4C overexcitation limiter	IEEE	Ref: [6], Sec. 10.7
oel_IEEE_OEL5C	IEEE Type OEL5C overexcitation limiter	IEEE	Ref: [6], Sec. 10.8
oel_PSSE_OEL5CU1	IEEE Type OEL5C overexcitation limiter	PSS/E	Ref: [7], Sec. 5.5
oel_CGMES_3_1_OverexcLim2	Overexcitation limiter	CGMES 3.1	Ref: [8], Sec. 8.4.10.4
oel_CGMES_3_1_OverexcLimIEEEOEL2C	IEEE Type OEL2C overexcitation limiter	CGMES 3.1	Ref: [8], Sec. 8.4.10.10 and [6], Sec. 10.5
oel_CGMES_3_1_OverexcLimX1	Maximum Excitation Limiter	CGMES 3.1	Ref: [8], Sec. 8.4.10.5

Type Name	Description	Library	Reference
oel_CGMES_3_1_OverexcLimX2	Maximum Excitation Limiter	CGMES 3.1	Ref: [8], Sec. 8.4.10.6
oel_X1	Field voltage over excitation limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.10.5
oel_X2	Field voltage or current overexcitation limiter	CGMES 2.4.15	Ref: [5], Sec. 9.6.10.6

Table 2.9: Overexcitation Limiter Models

2.10 Stator Current Limiter Models

Type Name	Description	Library	Reference
scl_IEEE_SCL1C	IEEE Type SCL1C stator current limiter	IEEE	Ref: [6], Sec. 12.2
scl_IEEE_SCL2C	IEEE Type SCL2C stator current limiter	IEEE	Ref: [6], Sec. 12.3

Table 2.10: Stator Current Limiter Models

2.11 PF/VAR Control Models

Type Name	Description	Library	Reference
pfq_IEEE_PF1	IEEE power factor controller Type 1	IEEE	Ref: [6], Sec. 13.4
pfq_CGMES_3_1_IEEEPF1	IEEE power factor controller Type 1	CGMES 3.1	Ref: [8], Sec. 8.4.14.6 and [6], Sec. 13.4
pfq_IEEE_PF2	IEEE power factor controller Type 2	IEEE	Ref: [6], Sec. 13.6
pfq_CGMES_3_1_IEEEPF2	IEEE power factor controller Type 2	CGMES 3.1	Ref: [8], Sec. 8.4.15.7 and [6], Sec. 13.6
pfq_CGMES_3_1_-Type2Common1	Power factor or VAr controller type 1	CGMES 3.1	Ref: [8], Sec. 8.4.15.5
pfq_IEEE_VAR1	IEEE var controller Type 1	IEEE	Ref: [6], Sec. 13.5
pfq_CGMES_3_1_IEEEVAR1	IEEE var controller Type 1	CGMES 3.1	Ref: [8], Sec. 8.4.14.5 and [6], Sec. 13.5
pfq_IEEE_VAR2	IEEE var controller Type 2	IEEE	Ref: [6], Sec. 13.7
pfq_CGMES_3_1_IEEEVAR2	IEEE var controller Type 2	CGMES 3.1	Ref: [8], Sec. 8.4.15.6 and [6], Sec. 13.7
vad_IEEE_1	IEEE voltage adjuster for Type 1 pf/var controllers	IEEE	Ref: [6], Sec. 13.3
vad_CGMES_3_1_IEEE1	IEEE voltage adjuster for Type 1 pf/var controllers	CGMES 3.1	Ref: [8], Sec. 8.4.16.2 and [6], Sec. 13.3
vad_IEEE_2	IEEE voltage adjuster for Type 2 pf/var controllers	IEEE	Ref: [6], Sec. 13.6 and Sec. 13.7
vad_CGMES_3_1_IEEE2	IEEE voltage adjuster for Type 2 pf/var controllers	CGMES 3.1	Ref: [8], [6], Sec. 13.6 and Sec. 13.7.

Type Name	Description	Library	Reference

Table 2.11: PF/VAR Control Models

2.12 Turbine/Engine Load Control Models

Type Name	Description	Library	Reference
tlc_IEEE_LCFB1	Turbine load controller model	IEEE	Ref: [3], Sec. 2.2
tlc_CGMES_3_1_LCFB1	Turbine load controller model	CGMES 3.1	Ref: [8], Sec. 8.4.7.3
tlc_PSSE_LCFB1	Turbine Load controller Model	PSS/E	Ref: [7], Sec. 8.1
tlc_TurbLCFB1	Turbine Load Controller	CGMES 2.4.15	Ref: [5], Sec. 9.6.7.3

Table 2.12: Turbine/Engine Load Control Models

3 Frames for Synchronous Generator Power Plant Dynamic Models

The *DIgSILENT* Library includes composite model frame types for synchronous generator controls, which can allocate any of the models from the previous sections. Sub-frames to be used in special cases are also provided in the *DIgSILENT* Library:

Frame Name	Description
Frame Single-Phase Synchronous Generator	Frame for single-phase synchronous generator power plant.
Frame Synchronous Generator	General frame for synchronous generator with standard library controls.
Sub-Frame CCPP HRSG/ST	Sub-Frame for a heat recovery steam generator (HRSG) unit of a combined cycle power plant (CCPP).
Sub-Frame PSS/DEC	Sub-frame to include a dec_* model in series with a pss_* model.
Sub-Frame Turbine and Governor	Sub-frame to represent the governor/turbine system by means of two different models.

Table 3.1: *DIgSILENT* Library Composite Model Frames for Synchronous Generator Power Plants

4 Generic Arrangements

If no specific information about the controllers is available for a given particular application, generic configurations may be used.

Steam Plant Model The steam turbine/governor models in the *DIgSILENT* library are used for independent steam turbines (non-combined cycle). If no detailed information is available, the standard thermal model IEEESGO may be used for the governor, enabling as many stages (HP, MP, LP) as required. The excitation system for this kind of unit is predominantly of the AC or static type, except for old installations, where DC systems may still be in operation.

Hydro Turbine Model The most commonly used hydro model is the *gov_CGMES_3_1_Hydro1* (equivalent to the PSS/E compatible HYGOV model) which represents the main dynamic effects of the energy source and prime mover. Static excitation systems are commonly used for modern hydro units.

Internal Combustion Model A diesel or gas engine unit can be represented by the DEGOVs (or a specially parameterized GGOV1 model). AC excitation systems are commonly used for these units.

Gas Turbine Model Gas units, governor/turbines are represented in many cases by GASTs and GGOVs models. The gas turbine/governor models can be used both for independent units or for units part of a combined cycle. The excitation system is predominately of the AC or static type.

5 References

[1] IEEE Recommended Practice for Excitation System Models for Power System Stability Studies (421.5 - 1992), 1992.

[2] IEEE Recommended Practice for Excitation System Models for Power System Stability Studies (421.5 - 2005), 2005.

[3] IEEE Technical Report PES-TR1. Dynamic Models for Turbine Governors in Power Systems Studies. Technical report, IEEE, January 2013.

[4] Model Library, PSS/E 33.5, 2013.

[5] Detailed Description of the CGMES Profiles. Version 2.4.15. Common Grid Model Exchange Standard, 2014.

[6] IEEE Recommended Practice for Excitation System Models for Power System Stability Studies (421.5 - 2016), 2016.

[7] Model Library, PSS/E 35.2, 2020.

[8] Draft IEC 61970 Energy Management System Application Program Interface (EMS-API) - Part 457: Common Information Model (CIM) for Dynamic Profile. Edition 2.00, 2022.